# Three-Player, Three-Level Benchmark with Inequalities as the Innermost Preference

This is the exact problem solved by `test/quasi_vs_reduced_preference_inequality.jl`. It is built in two steps:

1. The call

   `build_benchmark_problem(; num_players = 3, levels = 3, kind)`

   in `test/benchmark_problems.jl` constructs a three-level generalized Nash problem with *hard* coupled inequality constraints.

2. The helper `innermost_preference_inequality_goop` then deletes those hard constraints and re-attaches each player's inequality vector as a *fourth, highest-priority preference level*, flagged with

   `is_prioritized_constraint = true`.

   This is the encoding that `experiments/robotic_arm_core.jl` uses.

Everything below is fully expanded — no symbolic shorthand for the constants.

## 1. Notation

There are three players. Player $p \in \{1, 2, 3\}$ owns a decision vector

$$x_p = (x_{p,1},\, x_{p,2},\, x_{p,3},\, x_{p,4},\, x_{p,5}) \in \mathbb{R}^5, \qquad x = (x_1, x_2, x_3) \in \mathbb{R}^{15}.$$

In code, $x_{p,j}$ is `x[Block(p)][j]`. Write $x_{-p}$ for the other two players' vectors, which player $p$ treats as fixed parameters — the three player problems hold simultaneously, i.e. this is a generalized Nash equilibrium problem. There are parameters $\theta \in \mathbb{R}^3$, but every objective and constraint below ignores them; they are identically zero in the test.

**Preference ordering.** Preferences are stored `[outermost, ..., innermost]`, and the *innermost is the highest priority.* Player $p$ has four levels:

| level $\ell$ | controls | role | priority |
|---|---|---|---|
| 1 | $x_{p,5}$ | objective | lowest |
| 2 | $x_{p,3}$, $x_{p,4}$ | objective | |
| 3 | $x_{p,1}$, $x_{p,2}$ | objective | |
| 4 | $h_p(x) \geq 0$ | prioritized constraint | **highest** |

Each level's objective still ranges over the *whole* vector $x_p$; the coordinate lists say which coordinates that level's objective actually depends on.

## 2. The prioritized-constraint level

A level flagged `is_prioritized_constraint` does *not* enter as a constraint. It enters as the smooth penalty `ReducedGOOP.smooth_piecewise_preference_objective`, applied elementwise and summed. In code that function is `ifelse(h ≥ 0, 0.0, (0 - h)^(level + 2))`, i.e.

$$\varphi(h,\ell) = \begin{cases} 0, & h \geq 0, \\ (-h)^{\ell+2}, & h < 0. \end{cases}$$

The inequality sits at $\ell = 4$, so the exponent is $\ell + 2 = 6$:

$$P_p(x) \;=\; \sum_{m=1}^{4} \varphi\big(h_{p,m}(x),\, 4\big) \;=\; \sum_{m=1}^{4} \max\big(-h_{p,m}(x),\, 0\big)^6.$$

We have $P_p \geq 0$ always, and $P_p(x) = 0$ *iff* every $h_{p,m}(x) \geq 0$. So this level's argmin set is exactly the feasible set of the original hard constraints, whenever that set is nonempty.

## 3. The nested problem

Player $p$ solves the outermost level subject to being optimal for every higher-priority level:

$$\begin{aligned} \min_{x_p \in \mathbb{R}^5} \quad & J_p^{(1)}(x_p) \\ \text{subject to} \quad & x_p \in \mathcal{S}_p^{(2)}(x_{-p}), \end{aligned}$$

where each level's solution set is defined by the next one down:

$$\begin{aligned} \mathcal{S}_p^{(2)}(x_{-p}) &= \arg\min_{y \in \mathbb{R}^5} \; J_p^{(2)}(y) \quad \text{subject to} \quad y \in \mathcal{S}_p^{(3)}(x_{-p}), \\ \mathcal{S}_p^{(3)}(x_{-p}) &= \arg\min_{y \in \mathbb{R}^5} \; J_p^{(3)}(y) \quad \text{subject to} \quad y \in \mathcal{S}_p^{(4)}(x_{-p}), \\ \mathcal{S}_p^{(4)}(x_{-p}) &= \arg\min_{y \in \mathbb{R}^5} \; P_p(y, x_{-p}). \end{aligned}$$

The innermost problem is *unconstrained* — that is the whole point of the transformation. There are no hard constraints anywhere in this problem.

## 4. Coupling and constraint data

The coupled resource consumed by player $p$ in coordinate $j$ is

$$c_{p,j}(x) \;=\; x_{p,j} \;+\; 0.25 \sum_{k \neq p} x_{k,j},$$

with coupling weight 0.25 (`COUPLING_WEIGHT`). Each player has four inequality rows:

$$h_p(x) = \begin{pmatrix} C_{p,1} - c_{p,1}(x) \\ C_{p,2} - c_{p,2}(x) \\ c_{p,3}(x) - F_{p,3} \\ c_{p,4}(x) - F_{p,4} \end{pmatrix} \geq 0.$$

Rows 1 and 3 are *active* at the designed solution; rows 2 and 4 carry slack 1.

| $p$ | $C_{p,1}$ | $C_{p,2}$ | $F_{p,3}$ | $F_{p,4}$ |
|---|---|---|---|---|
| 1 | 0.6375 | 2.4250 | 0.8625 | 0.8750 |
| 2 | 0.6750 | 2.5000 | 0.9000 | 0.9500 |
| 3 | 0.7125 | 2.5750 | 0.9375 | 1.0250 |

| $p$ | $t_{p,1}$ | $t_{p,2}$ | $t_{p,3}$ | $t_{p,4}$ | $t_{p,5}$ |
|---|---|---|---|---|---|
| 1 | 1.00 | 0.90 | 0.15 | 1.20 | 0.80 |
| 2 | 1.05 | 1.00 | 0.20 | 1.30 | 0.90 |
| 3 | 1.10 | 1.10 | 0.25 | 1.40 | 1.00 |

The right-hand table lists the unconstrained objective targets $t_p$ — the point each objective would pick with no constraints.

## 5. `kind = :quadratic`, fully expanded

Objectives are squared distances to the targets; the constraint rows are affine.

**Player 1**

$$\begin{aligned}
&\min_{x_1 \in \mathbb{R}^5} \quad (x_{1,5} - 0.80)^2 \\
&\text{subject to} \quad x_1 \in \mathcal{S}_1^{(2)}(x_2, x_3) \\
\mathcal{S}_1^{(2)} &= \arg\min_y \ (y_3 - 0.15)^2 + (y_4 - 1.20)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_1^{(3)}, \\
\mathcal{S}_1^{(3)} &= \arg\min_y \ (y_1 - 1.00)^2 + (y_2 - 0.90)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_1^{(4)}, \\
\mathcal{S}_1^{(4)} &= \arg\min_y \ \Big[\max\big(-(0.6375 - y_1 - 0.25x_{2,1} - 0.25x_{3,1}),\, 0\big)^6 \\
&\qquad + \max\big(-(2.4250 - y_2 - 0.25x_{2,2} - 0.25x_{3,2}),\, 0\big)^6 \\
&\qquad + \max\big(-(y_3 + 0.25x_{2,3} + 0.25x_{3,3} - 0.8625),\, 0\big)^6 \\
&\qquad + \max\big(-(y_4 + 0.25x_{2,4} + 0.25x_{3,4} - 0.8750),\, 0\big)^6\Big].
\end{aligned}$$

**Player 2**

$$\begin{aligned}
&\min_{x_2 \in \mathbb{R}^5} \quad (x_{2,5} - 0.90)^2 \\
&\text{subject to} \quad x_2 \in \mathcal{S}_2^{(2)}(x_1, x_3) \\
\mathcal{S}_2^{(2)} &= \arg\min_y \ (y_3 - 0.20)^2 + (y_4 - 1.30)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_2^{(3)}, \\
\mathcal{S}_2^{(3)} &= \arg\min_y \ (y_1 - 1.05)^2 + (y_2 - 1.00)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_2^{(4)}, \\
\mathcal{S}_2^{(4)} &= \arg\min_y \ \Big[\max\big(-(0.6750 - y_1 - 0.25x_{1,1} - 0.25x_{3,1}),\, 0\big)^6 \\
&\qquad + \max\big(-(2.5000 - y_2 - 0.25x_{1,2} - 0.25x_{3,2}),\, 0\big)^6 \\
&\qquad + \max\big(-(y_3 + 0.25x_{1,3} + 0.25x_{3,3} - 0.9000),\, 0\big)^6 \\
&\qquad + \max\big(-(y_4 + 0.25x_{1,4} + 0.25x_{3,4} - 0.9500),\, 0\big)^6\Big].
\end{aligned}$$

**Player 3**

$$\begin{aligned}
&\min_{x_3 \in \mathbb{R}^5} \quad (x_{3,5} - 1.00)^2 \\
&\text{subject to} \quad x_3 \in \mathcal{S}_3^{(2)}(x_1, x_2) \\
\mathcal{S}_3^{(2)} &= \arg\min_y \ (y_3 - 0.25)^2 + (y_4 - 1.40)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_3^{(3)}, \\
\mathcal{S}_3^{(3)} &= \arg\min_y \ (y_1 - 1.10)^2 + (y_2 - 1.10)^2 \quad \text{s.t.} \quad y \in \mathcal{S}_3^{(4)}, \\
\mathcal{S}_3^{(4)} &= \arg\min_y \ \Big[\max\big(-(0.7125 - y_1 - 0.25x_{1,1} - 0.25x_{2,1}),\, 0\big)^6 \\
&\qquad + \max\big(-(2.5750 - y_2 - 0.25x_{1,2} - 0.25x_{2,2}),\, 0\big)^6 \\
&\qquad + \max\big(-(y_3 + 0.25x_{1,3} + 0.25x_{2,3} - 0.9375),\, 0\big)^6 \\
&\qquad + \max\big(-(y_4 + 0.25x_{1,4} + 0.25x_{2,4} - 1.0250),\, 0\big)^6\Big].
\end{aligned}$$

## 6. `kind = :nonlinear`, fully expanded

Same structure, two substitutions:

- every squared term $(u - t)^2$ becomes $\cosh(u - t) - 1$;
- every constraint row $h$ becomes $e^h - 1$.

The second substitution preserves the feasible set and the active set exactly, since $e^r - 1 = 0 \iff r = 0$ and $e^r - 1 > 0 \iff r > 0$. It does change the *geometry*: the rows are no longer affine, and cosh has nonvanishing derivatives of every order. Note $-(e^h - 1) = 1 - e^h$, which is how the penalty is written below.

**Player 1**

$$\begin{aligned}
&\min_{x_1 \in \mathbb{R}^5} \quad \cosh(x_{1,5} - 0.80) - 1 \\
&\text{subject to} \quad x_1 \in \mathcal{S}_1^{(2)}(x_2, x_3) \\
\mathcal{S}_1^{(2)} &= \arg\min_y \ \big[\cosh(y_3 - 0.15) - 1\big] + \big[\cosh(y_4 - 1.20) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_1^{(3)}, \\
\mathcal{S}_1^{(3)} &= \arg\min_y \ \big[\cosh(y_1 - 1.00) - 1\big] + \big[\cosh(y_2 - 0.90) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_1^{(4)}, \\
\mathcal{S}_1^{(4)} &= \arg\min_y \ \Big[\max\big(1 - e^{\,0.6375 - y_1 - 0.25x_{2,1} - 0.25x_{3,1}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,2.4250 - y_2 - 0.25x_{2,2} - 0.25x_{3,2}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_3 + 0.25x_{2,3} + 0.25x_{3,3} - 0.8625},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_4 + 0.25x_{2,4} + 0.25x_{3,4} - 0.8750},\, 0\big)^6\Big].
\end{aligned}$$

**Player 2**

$$\begin{aligned}
&\min_{x_2 \in \mathbb{R}^5} \quad \cosh(x_{2,5} - 0.90) - 1 \\
&\text{subject to} \quad x_2 \in \mathcal{S}_2^{(2)}(x_1, x_3)
\end{aligned}$$

$$
\begin{aligned}
\mathcal{S}_2^{(2)} &= \arg\min_y \ \big[\cosh(y_3 - 0.20) - 1\big] + \big[\cosh(y_4 - 1.30) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_2^{(3)}, \\
\mathcal{S}_2^{(3)} &= \arg\min_y \ \big[\cosh(y_1 - 1.05) - 1\big] + \big[\cosh(y_2 - 1.00) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_2^{(4)}, \\
\mathcal{S}_2^{(4)} &= \arg\min_y \ \Big[\max\big(1 - e^{\,0.6750 - y_1 - 0.25x_{1,1} - 0.25x_{3,1}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,2.5000 - y_2 - 0.25x_{1,2} - 0.25x_{3,2}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_3 + 0.25x_{1,3} + 0.25x_{3,3} - 0.9000},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_4 + 0.25x_{1,4} + 0.25x_{3,4} - 0.9500},\, 0\big)^6\Big].
\end{aligned}
$$

**Player 3**

$$
\begin{aligned}
&\min_{x_3 \in \mathbb{R}^5} \quad \cosh(x_{3,5} - 1.00) - 1 \\
&\text{subject to} \quad x_3 \in \mathcal{S}_3^{(2)}(x_1, x_2)
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{S}_3^{(2)} &= \arg\min_y \ \big[\cosh(y_3 - 0.25) - 1\big] + \big[\cosh(y_4 - 1.40) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_3^{(3)}, \\
\mathcal{S}_3^{(3)} &= \arg\min_y \ \big[\cosh(y_1 - 1.10) - 1\big] + \big[\cosh(y_2 - 1.10) - 1\big] \quad \text{s.t.} \quad y \in \mathcal{S}_3^{(4)}, \\
\mathcal{S}_3^{(4)} &= \arg\min_y \ \Big[\max\big(1 - e^{\,0.7125 - y_1 - 0.25x_{1,1} - 0.25x_{2,1}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,2.5750 - y_2 - 0.25x_{1,2} - 0.25x_{2,2}},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_3 + 0.25x_{1,3} + 0.25x_{2,3} - 0.9375},\, 0\big)^6 \\
&\qquad + \max\big(1 - e^{\,y_4 + 0.25x_{1,4} + 0.25x_{2,4} - 1.0250},\, 0\big)^6\Big].
\end{aligned}
$$

## 7. Ground truth

The family is constructed so that the hard-constrained problem has the closed-form solution

$$
x_1^\star = \begin{pmatrix} 0.40 \\ 0.90 \\ 0.55 \\ 1.20 \\ 0.80 \end{pmatrix}, \qquad x_2^\star = \begin{pmatrix} 0.45 \\ 1.00 \\ 0.60 \\ 1.30 \\ 0.90 \end{pmatrix}, \qquad x_3^\star = \begin{pmatrix} 0.50 \\ 1.10 \\ 0.65 \\ 1.40 \\ 1.00 \end{pmatrix}.
$$

`case.expected` in the test is this vector, flattened to length 15.

**How it is constructed.** $t_p$ differs from $x_p^\star$ only in coordinates 1 and 3:

$$
t_{p,1} = x_{p,1}^\star + 0.6, \qquad t_{p,3} = x_{p,3}^\star - 0.4,
$$

with $t_{p,2} = x_{p,2}^\star$, $t_{p,4} = x_{p,4}^\star$, $t_{p,5} = x_{p,5}^\star$. Coordinates 2, 4 and 5 are therefore already at their unconstrained optimum and no constraint binds them. Coordinates 1 and 3 are pulled off target by exactly the two active rows: $C_{p,1}$ is *defined* as $c_{p,1}(x^\star)$ and $F_{p,3}$ as $c_{p,3}(x^\star)$, while the two inactive rows are offset by $\pm 1$.

**Verification at $x^\star$** (identical for all three players):

| row | $h_{p,m}(x^\star)$, `:quadratic` | $e^h - 1$, `:nonlinear` | status |
|---|---|---|---|
| 1 | 0 | 0 | active |
| 2 | 1 | $e - 1 \approx 1.71828$ | inactive |
| 3 | 0 | 0 | active |
| 4 | 1 | $e - 1 \approx 1.71828$ | inactive |

Hence $P_p(x^\star) = 0$ for every player: $x^\star$ attains the exact minimum of the highest-priority level. Since that level's argmin set equals the feasible set, the nested problem above has the *same* solution as the hard-constrained original — in exact arithmetic.

The $3 \times 3$ coupling matrix has 1 on the diagonal and 0.25 off it, so it is nonsingular and the two active rows pin each player's share individually, not just the aggregate.

## 8. Why solves do not land on $x^\star$

$x^\star$ sits exactly on the boundary of the two active rows, where $h_{p,1} = h_{p,3} = 0$. The penalty $\max(-h, 0)^6$ has value *and first five derivatives all zero* there. The highest-priority level therefore exerts no gradient at the solution: nothing pulls an iterate back onto the active constraints, and the lower-priority objectives — which want coordinate 1 larger and coordinate 3 smaller — push it into mild infeasibility until the sixth power grows enough to resist.

Measured with the test's configuration (KLU, backtracking, tol $= 10^{-4}$, `max_outer_iters = 1`):

| `kind` | $z_0$ | $\|x - x^\star\|_\infty$ reduced / quasi | max violation |
|---|---|---|---|
| `:quadratic` | `expected` | 0.06068 / 0.06068 | 0.0910 |
| `:quadratic` | `zeros` | 0.72728 / 0.72728 | 0.0909 |
| `:quadratic` | `offset` | 0.06057 / 0.06057 | 0.0909 |
| `:nonlinear` | `expected` | 0.06257 / 0.06257 | 0.0896 |
| `:nonlinear` | `zeros` | 0.73006 / 0.73006 | 0.0907 |
| `:nonlinear` | `offset` | 0.06240 / 0.06240 | 0.0894 |

The residual is not what limits accuracy here — the degenerate penalty is. Tightening the tolerance to $10^{-6}$ makes every solve report `:failed` (the iteration stalls at $\|F\| \approx 1.5 \times 10^{-5} \ldots 2.9 \times 10^{-5}$) yet moves the answer only from 0.061 to 0.046 away from $x^\star$, with the violation shrinking from 0.091 to 0.069.

This gap is a property of the $(-h)^6$ encoding, not of the `:reduced` / `:quasi` choice: the two formulations agree with each other to between $3 \times 10^{-15}$ and $4 \times 10^{-6}$, i.e. four to nine orders of magnitude closer to each other than either is to $x^\star$.

It is the same encoding `robotic_arm_core.jl` uses for its collision and speed constraints, so the same slack should be expected there — a violation of $10^{-2}$ registers as a penalty of only $10^{-12}$.